

Warm-up

Of the two series below, the ratio test works better on one and the root test works better on the other. Determine which series is which and analyze the convergence.

$$(1) \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^n}, \quad (2) \sum_{n=1}^{\infty} \frac{(-1)^n 3^n}{5^n + 1}.$$

Medium

Analyze the convergence of the following series:

$$(1) \sum_{n=1}^{\infty} \frac{2^n}{n!}, \quad (2) \sum_{n=1}^{\infty} \frac{(-2)^n}{n^3 + n}, \quad (3) \sum_{n=1}^{\infty} \frac{(-1)^n (3n^2 + 5n + 2)^n}{(5n^2 - 3)^n}, \quad (4) \sum_{n=1}^{\infty} \frac{n^2 + n - 3}{n^2 4^n}.$$

Stretch

(1) Use the ratio test to show that $\sum_{n=1}^{\infty} \frac{a^n n!}{n^n}$ converges for $a < e$ and diverges for $a > e$.

(2) The ratio test is inconclusive if $a = e$. By comparing areas,

$$\ln(n!) = \ln(1) + \ln(2) + \cdots + \ln(n) > \int_1^n \ln(x) dx.$$

Evaluate this integral to show that $n! > n^n / e^{n-1}$. Does the series in (1) converge or diverge if $a = e$?